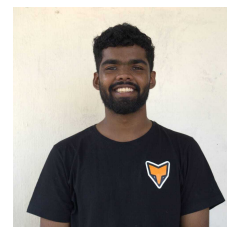


KAUSTUBH PANDEY

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[kpkaustubhkp.github.io](https://github.com/kpkaustubhkp)



Education

Manipal Institute of Technology

B.Tech in Electronics & Communication (VLSI)

– Current CGPA: 7.73 / 10 (as of 6th Semester)

2023 – Present

Manipal, Karnataka

Relevant Coursework

- | | | | |
|---------------------|--------------------|----------------------|-------------------------|
| • VLSI Design | • Analog IC Design | • Analog Circuits | • Semiconductor Physics |
| • Digital IC Design | • Physical Design | • FPGA System Design | • Verilog HDL |

Experience

VLSI System Design – PCB Design Intern

Feb 2025 – Oct 2025

PCB Designer

Manipal, India

- Designed and laid out the evaluation board for the Thejas32 (by Vega Processor & CDAC) using KiCad, including schematic capture and PCB layout.
- Collaborated on creating a shield for their FPGA board PCB design, integrating interface components and ensuring signal integrity.

Project MANAS – Hardware Subsystem, Autonomous UGV Team

June 2024 – Present

Hardware Engineer

Manipal, India

- Designed power distribution boards with high-current switching and implemented current sensing using current-sense ICs.
- Led PCB design (KiCad) for motor/microcontroller interfaces; integrated CAN-bus transceivers.
- **3rd place AutoNav Challenge (IGVC 2025) and PIMA Winner (ISDC 2025)** with autonomous UGV & UAV platforms.

Projects

Low-Dropout Regulator (LDO) Design | Cadence Virtuoso, Calibre, Analog IC Design

Feb 2026 – June 2026

- Designed a 180nm CMOS LDO in Cadence Virtuoso regulating Vout to 1.2V, using an OTA-based error amplifier with resistive feedback and a bandgap reference.
- Completed full-custom layout with DRC/LVS verification in Calibre; validated closed-loop DC regulation via simulation.

FPGA Lane & Obstacle Detection | SystemVerilog, Vivado

Feb 2026 – June 2026

- Designed a 6-stage SystemVerilog RTL pipeline (grayscale conversion, Gaussian blur, edge detection, ROI filtering, Hough transform) for real-time lane detection on an Artix-7 FPGA

PID Motor Controller | Verilog, Vivado, Icarus Verilog, GTKWave

May 2025

- **Published in IEEE:** DOI: 11232594 - Digital PID Controller for DC Motor Speed Regulation on FPGA.
- Designed Verilog PID controller with encoder feedback; synthesized pipelined architecture on Vivado FPGA for deterministic loops.

4H-SiC Trench U-MOSFET with BPT | Synopsys Sentaurus TCAD, Device Physics

Oct 2025

- Co-authored research on 4H-SiC trench U-MOSFET achieving 556 V breakdown voltage using breakdown point transfer (BPT) technique.
- Performed 2D TCAD device simulations; optimized drift region and electric field profiles for high-voltage performance.

Technical Skills

Languages: SystemVerilog, Verilog, C, Python, Bash, Latex

Tools: KiCad, Vivado, Icarus Verilog, GTKWave, Cadence, Synopsys Sentaurus TCAD, LTSpice

Technologies: PCB Design, FPGA, Power Devices, Semiconductor Physics, Physical Design

Leadership / Extracurricular

IRC & ISDC 2026 (SPROS WEEK) Core Committee

Jan 2026 – Feb 2026

Core Committee Member

Manipal Institute of Technology

- Organizing IRC and ISDC 2026 events as part of SPROS WEEK core committee.
- Coordinating technical competitions, workshops, and student participation.